// README · MODULE DOCUMENTATION

Energy *Balance*

Aggregates building consumption and plant production to primary substation (cabin) level, computes hourly energy balance vectors, and derives annual and monthly KPIs — SCI, SSI, OPI — with environmental and economic metrics.

PostgreSQL 15 | SCI · SSI · OPI | Cabin & Province level

Regional scope: This documentation covers the **Sicilia** implementation. The identical SQL logic applies to **Puglia** and **Calabria** with schema substitution only.

FILE	LANGUAGE	PRIMARY OUTPUT
Energy_Balance_Sicilia/	SQL (PostgreSQL)	results_cabin · results_province

KPIS

SCI · SSI · OPI · CO_2 · €

Calendar Setup & Array Unnesting

1.1 **2024 FR/FS Day Count Calendar**

A temporary table `temp_days_in_month` stores the number of working days (FR) and festive/weekend days (FS) for each of the 12 months in 2024. This table is joined later to convert per-typical-day hourly values into monthly energy totals (kWh/month = kWh/typical-day × days-in-month).

2.1 **UNNEST Consumption Arrays → Cabin Hourly Demand**

The 24-element `total_kwh_arr` from `sicilia_building_hourly_consumption` is unnested using `CROSS JOIN LATERAL generate_series(1,24)`, producing one row per building × month × day-type × hour. Values are then **summed by cabin** (`cod_ac`) to get total cabin hourly demand.

```
CROSS JOIN LATERAL (
  SELECT gs.hour,
         COALESCE(h.total_kwh_arr[gs.hour], 0) AS total_kwh
  FROM generate_series(1, 24) AS gs(hour)
) AS hour_idx
```

3.1 **UNNEST Production Arrays → Cabin Hourly Supply**

The identical unnesting logic is applied to `sicilia_actual_plants_hourly_production`. Plant production arrays are unnested and summed by the cabin area (`cod_ac`) they belong to, giving total cabin hourly supply.

// STEPS 04 – 07

Balance Computation & KPI Derivation

4.1

Hourly Balance: SC, UD, OP

For each cabin × month × day-type × hour, three balance quantities are computed and immediately multiplied by the number of days in that period:

```
-- Self-Consumption (shared energy)
SC  = LEAST(demand, supply) × days

-- Unmet Demand (grid import)
UD  = GREATEST(demand − supply, 0) × days

-- Over-Production (grid export)
OP  = GREATEST(supply − demand, 0) × days
```

5.1

Annual KPIs — Cabin Level (results_cabin)

Hourly balance rows are aggregated by cabin. Two dimensionless indices are computed:

```
-- Self-Consumption Index: share of production consumed locally
SCI = SUM(sc_kwh) / SUM(tp_kwh)    ∈ [0, 1]

-- Self-Sufficiency Index: share of demand met locally
SSI = SUM(sc_kwh) / SUM(tc_kwh)    ∈ [0, 1]
```

NULL is returned instead of division-by-zero when the denominator is 0.

6.1

Province-Level Aggregation (results_province)

The same aggregation is performed at province level. Values are expressed in **GWh** (dividing by 1,000,000) for readability. SCI and SSI are computed from the province-level sums, not as averages of cabin-level indices.

7.1

Monthly KPIs — Cabin Level (results_cabin_monthly)

The same GROUP BY logic is applied with an additional `month` dimension, producing 12 rows per cabin with monthly SCI and SSI — useful for seasonality analysis.

Advanced KPIs, Environmental & Economic Metrics

9.1 Over-Production Index (OPI)

```
OPI = annual_op_kwh / annual_supply_kwh   ∈ [0, 1]
```

A high OPI signals that most generated energy cannot be absorbed locally — indicating either infrastructure oversizing or the need for storage.

9.2 Avoided CO_2 Emissions

Avoided CO_2 is calculated using Italy's average grid emission factor:

```
avoided_co2_tons = annual_sc_kwh × 0.256 / 1000
-- 0.256 kg CO₂eq / kWh (Italian grid average)
```

9.3 Economic Benefit Estimation

```
economic_benefit_€ = (sc_kwh × 0.22) + (op_kwh × 0.10)
-- 0.22 €/kWh for self-consumed energy (avoided purchase)
-- 0.10 €/kWh for over-produced/shared energy (incentive)
```

10.1 Environmental Engineering Nexus (Air & Water)

Province-level avoided pollutants and water savings replace the fossil-heavy national grid mix with local RES generation:

POLLUTANT	FACTOR	UNIT
CO_2 equivalent	256	tons / GWh
NO_x	0.4	tons / GWh
SO_2	0.15	tons / GWh
PM_{10}	0.02	tons / GWh
Water saved	2,500	m^3 / GWh

OUTPUT TABLES SUMMARY

`results_cabin`	Annual SCI, SSI, demand, supply, SC, UD, OP per cabin
`results_province`	Annual GWh-level KPIs aggregated by province
`results_cabin_monthly`	Monthly SCI, SSI per cabin (seasonality)
`advanced_results_cabin`	+ OPI, avoided CO_2 (tons), economic benefit (€)
`advanced_results_province`	Province-level OPI, CO_2 (kilotons), benefit (M€)
`environmental_nexus_province`	Avoided NO_x, SO_2, PM_{10}, water savings per province

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Note: The initial draft of this README documentation was generated with the assistance of Claude AI and subsequently reviewed, verified, and edited by the author for accuracy.